ABDULLAH AL MARUF

SOFTWARE ENGINEER

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Austin, TX, USA

PROFILE

Passionate Software Engineer with **7 years** of experience developing **Kubernetes** tools in **Go** and building cloud-native infrastructure using **Infrastructure as Code (IaC)** with **Terraform**, Helm, and ArgoCD across **AWS** and **Azure**. Skilled in backend development with **Java** and algorithmic problem solving in **C++**. Led the KubeDB project at AppsCode and contributed to open-source tools.

EXPERIENCE

Software Engineer II

Oct, 2022 - Present

ZEISS Medical Technology

Temple, TX, USA

- Collaborated with cross-functional teams to analyze and design high-level system architecture.
- Migrated software and infrastructure from AWS ECS to Kubernetes on **Azure AKS**, achieving **~3x faster** system performance.
- Designed and implemented infrastructure as code (IaC) with Terraform, **Helm**, and **ArgoCD**, with monitoring and alerting via **Prometheus** and Grafana.
- Refactored **backend** services in **nodejs** to seamlessly integrate with AWS and Azure resources, unifying the codebase for cloud-agnostic and scalable development.

Intern

Summer 2021, 2022

Red Hat

Brno, CZ (Remote)

- Led a research team to conduct a Systematic Mapping Study on technical debt resulting from architectural degradation and code smells.
- Analyzed issue tickets and version control data to identify and quantify code debt.
- Applied static analysis and process mining to detect architectural degradation in microservice systems.
- Technologies: **Java**, **Spring Boot,** Git, Static Analysis.

DevOps Engineer

Feb, 2020 - Dec, 2020

Grameen Digital Health

Dhaka, Bangladesh

- Designed and migrated **scalable** infrastructure to Kubernetes, ensuring high availability and resilience.
- Managed hybrid cloud infrastructure consisting of **AWS** and on-premises **OpenStack**, implementing automation pipelines for deployment and scaling.
- Optimized **CI/CD** workflows using Terraform, **Ansible**, **Jenkins**, and Docker, improving consistency.
- Enhanced system reliability and cost efficiency by implementing monitoring (Prometheus, Grafana) and improving auto-scaling (**~25% cost reduction**).

Software Engineer

May, 2017 - Jan, 2020

AppsCode Ltd.

Dhaka, Bangladesh

- Developed Kubernetes **controller** tools in **Go** and Docker to extend Kubernetes functionality.
- Designed and implemented KubeDB, a Kubernetes-native, cloud-agnostic database management tool built to operate consistently across GCP, AWS, and Azure Kubernetes clusters.
- Led a team of 5 to develop core KubeDB features and enhanced **MongoDB**, **Elasticsearch**, and **Postgresql** clustering.
- Scaled KubeDB community engagement via Slack and meetups, boosting returning users by **200%.**
- Improved database reliability by implementing automated backup and recovery workflows using Stash.

EDUCATION

Baylor University

M.S. in Computer Science

Chittagong University of Engineering & Technology

B.Sc. in Computer Science & Engineering

PROUD OF

Competitive Programming

- Solved **1100+** problems on different Online Judges.
- **Divisional Champion** of ACM ICPC Dhaka Regional '14 & '15.

Open Source Participation

User, follower, and contributor.

Research Impact

- **600+** citations across peer-reviewed publications.
- Focus: microservice architecture, observability, and system design.

KubeDB's growth

From very few to thousands of daily users.

Leadership

Served as President of Cuet Computer Club. Also acted as team lead of KubeDB.

FIND ME ONLINE

- https://the-redback.com
- linkedin: maruftuhin
- github: the-redback
- stackoverflow: 4628962
- scholar: Prun7T8AAAAJ
- codeforces: the_redback

TECHNICAL SKILLS

GO, Java, C++, Python, Terraform, Bash, Linux

Kubernetes, Docker, Helm, ArgoCD, Ansible, Jenkins

Azure, AWS, GCP, EKS, AKS, GKE, Linkerd, Istio

PostgreSQL, MySQL, MongoDB, Kafka, RabbitMQ, CI/CD

Prometheus, Grafana, System Design, Cloud Native, Git

TOP PROJECTS

KubeDB

https://kubedb.com

Kubernetes controller powering a database management platform, automating provisioning, scaling, backup, recovery, and failover.

- Led the team to design & implement the core features.
- Designed and implemented MongoDB clustering system support.
- Improved PostgreSQL and Elasticsearch clustering system.
- Worked with Prometheus exporters to export database metrics.

Stash

https://stash.run

Kubernetes native disaster recovery solution (uses **restic**) for backup and restore volumes and databases in Kubernetes.

- Improved database backup system using stash.
- Designed and implemented Elasticsearch and MongoDB cluster backup system from scratch.
- Integrated Stash support within Kubedb to backup/restore dbs.

Microservice Architecture (MSA) Reconstruction

- Developed a system for **software architecture** reconstruction (SAR) from distributed tracing logs using **Istio** and **Java**.
- Leveraged microservice **telemetry data** for dynamic analysis to identify runtime anti-patterns and architectural drift.
- Generated automated visual graphs (Graphviz) to assist in architecture evolution tracking and system auditing.

SELECTED PUBLICATIONS

- "Microservice Architecture Reconstruction and Visualization Techniques: A Review." IEEE SOSE, 2022. (110 citations)
- "On Microservice Analysis and Architecture Evolution: A Systematic Mapping Study." Applied Sciences, 2021. (106 citations)
- "Catalog and Detection Techniques of Microservice Anti-Patterns and Bad Smells: A Tertiary Study." Journal of Systems and Software, 2023. (63 citations)
- "Using Microservice Telemetry Data for System Dynamic Analysis." IEEE SOSE, 2022. (56 citations)
- Full list: *Google Scholar* (600+ citations)